1 What is meant by steady state response?

Ans: Steady State Response refers to the behaviour of a system after it has been given enough time to stabilize in response to a constant input or disturbance.

2 What do you mean by the concept of Impedance and Phase angle?

Resistance restricts the flow of current by opposing free electron movement. Each element has some resistance; for example, an inductor has some resistance; a capacitance also has some resistance. Impedance is a complex quantity having real and imaginary parts; where the real part is the resistance and the imaginary part is the reactance of the circuit.

3 What is complex impedance?

Complex impedance is the total opposition offered by the circuit elements to ac

Current, and can be displayed on the complex plane. The impedance is denoted by

Z. Here the resistance R is the real part of the impedance, and the reactance XL is

the imaginary part of the impedance. The resistance R is located on the real axis.

The inductive reactance XL is located on the positive j axis. The resultant of R

and XL is called the complex impedance.

4) Define the terms Impedance and Phase angle.

Resistance restricts the flow of current by opposing free electron movement.

The angle between impedance and resistance is the phase angle between the current and voltage applied to the circuit

5) Write the expression for total impedance of the circuit having (i) Only Resistance (ii) Pure Inductor

(iii) Pure Capacitor (iv) R-L parameters (v) R-C parameters (vi) R-L-C parameters. Write theexpression for phase difference in all the above cases.

Ans: R-C parameters:

Form Fig. 5.4, impedance $Z = \sqrt{R^2 + X_C^2}$ or $\sqrt{R^2 + (1/\omega C)^2}$ and angle $\theta = \tan^{-1}(1/\omega CR)$. Here, the impedance, Z, is the vector sum of resistance and capacitive reactance.The angle between resistance and impedance is the phase angle between the applied voltage and current in the circuit.

R-L parameters:

Figure 5.2 is called the impedance diagram for the RL circuit. From Fig. 5.2, the impedance $Z = \sqrt{R^2 + (\omega L)^2}$, and angle $\theta = \tan^{-1} \omega L/R$.

RLC parametrs

of the two reactance. The impedance for an RLC series circuit is given by $Z = \sqrt{R^2 + (X_L - X_C)^2}$. Similarly, the phase angle for an RLC circuit is

$$\theta = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$

6 Why an impedance represented by a complex number? How is complex impedance dependent onfrequency?

Ans : In complex impedance the real part consists of a resistor which dissipates active power (heat) , whereas the imaginary part consists of inductive reactance or capacitive reactance or both which is frequency dependent and is responsible for the reactive power in the circuit. Reactive power VAR doesn't do any real work

7 Define the term 'Time Constant'. and write formulae for RL, RC Circuit.

Ans:

T=L/R T=1/RC

11 What is Star and Delta connection?

In a Star Connection, the system has a neutral wire, and the line voltage is root three times the phase voltage. It is a balanced circuit connection. In a Delta Connection, there is no neutral wire, and the line voltage is equal to the phase voltage.

12 What is the purpose of Star-Delta conversion?

It is often useful to transform delta network to star and star network to delta while solving an electrical network. The transformation can make the calculations simpler and easier. Some resistor networks cannot be simplified using the usual series and parallel combinations.

13 If P is the power of a star connected system, then what will be the power of an equivalent delta connected system?

Ans:3P